

Jose Andres Melgoza Mendoza

Embedded Systems and FPGA Design Engineer

📍 Madrid, Spain ✉ andres.mel.men@gmail.com ☎ +34 623 78 3344

Summary

Embedded Systems and FPGA Engineer with 7+ years of experience in software development, hardware acceleration, and FPGA design/verification for safety-critical and space-grade systems. Extensive hands-on experience in C for real-time embedded software, C++ for modular architectures, and Python for automation and tool integration. Experienced with AUTOSAR, ISO 26262, ASPICE, and ECSS-aligned workflows, with strong capabilities in hardware/software co-design, performance optimization, and robust validation in constrained environments. Skilled in MCAL, BSW, and complex driver integration in automotive ECUs, and in cross-functional debugging and system-level bring-up. Focused on embedded vision, real-time systems, and dependable low-level engineering.

Technical Skills

Languages: C, C++, Python, VHDL, Bash, SQL

MCUs/SoCs: STM32, MSP430, ARM Cortex-M, Xilinx Zynq, Artix-7

RTOS: FreeRTOS, Zephyr, CMSIS-RTOS

Tools: Vivado, CubeIDE, Keil, IAR, Altium Designer, STM32CubeMX, ModelSim, QuestaSim, Sigasi, Synplify, CANoe, Lauterbach, J-Link, VectorCAST, Coverity, MISRA tools

Protocols: SPI, I2C, UART, CAN, LIN

Linux: Embedded Linux, Yocto, Buildroot, U-Boot, Device Tree

Build Systems: Make, CMake

Advanced: REST API integration, Protocol Buffers

Verification: VHDL testbenches, QuestaSim/ModelSim, UCDB coverage, SystemVerilog Assertions

Requirements & Standards: ECSS standards, IBM DOORS, SUMO requirement tracing

Team/DevOps: Git, GitLab CI, Jenkins, Jira, IBM Rational Team Concert (RTC)

Documentation: LaTeX, IBM DOORS, R4J

Experience

Exceltic, FPGA Design Engineer (Thales Alenia Space – Inmarsat-8 Project)

- Implemented VHDL modules for ADC housekeeping (HK), including telemetry acquisition, monitoring, and reporting.
- Designed PLL scrubbing logic to detect and recover from lock issues for satellite platforms.
- Built unit-level testbenches and simulation checks to validate new VHDL blocks before integration.

Madrid, Spain
Feb 2026 – Present



Exceltic, FPGA Verification Engineer (Thales Alenia Space – CTEG Project)

- Verification engineer for the CTEG (Configurable Telecommunication Gateway), a space-grade FPGA-based secure router for satellite communications.
- Developed and executed VHDL testbenches to validate telecommand/telemetry flows, ensuring encryption/decryption compliance with ESU interfaces.
- Performed coverage analysis (QuestaSim/UCDB) and created requirement traceability matrices (DOORS, SUMO) to guarantee full functional verification of FPGA requirements.
- Debugged timing, protocol, and error-handling scenarios (timeouts, busy signals, authentication failures, bit errors) at simulation level.
- Coordinated with cross-site teams (Spain, France, Italy) to align design changes with ECSS standards and project milestones.

Madrid, Spain
May 2025 – Jan 2026



Visteon VTCQ, Senior DiApps & BMS Developer

- Developed DiApps modules in C for vehicle HMI, including menu navigation, icon rendering, and display personalization based on vehicle state.
- Programmed interaction logic between physical controls (buttons, rotary knobs) and software input handlers using C.
- Maintained and extended BMS communication layers using UDS over CAN, with diagnostics and status reporting logic.
- Wrote and reviewed ASPICE-compliant documentation: SRD, SRS, LLD, HLD.
- Debugged embedded software with Lauterbach (TRACE32) and Vector CANoe across multiple integration stages.
- Participated in software/hardware integration events (SW/HW Bring-up) for pre-production units.
- Worked with cross-functional teams to define ARXML interfaces and signal mappings using Vector tools.
- Collaborated on low-level software configuration using DaVinci Configurator/Developer.

Querétaro, Mexico
Nov 2021 – Sep 2024

Visteon

Samsung Electronics, Automation & Vision Systems Engineer

- Designed machine vision systems using OpenCV in C++ and Python to detect defects in real-time during manufacturing.
- Developed FPGA-based hardware acceleration for convolutional filters, offloading image preprocessing tasks.
- Trained lightweight convolutional neural networks and deployed them using fixed-point logic in VHDL.
- Integrated HMIs and PLCs to control robotic test benches based on visual classification outcomes.
- Validated end-to-end pipeline from image capture to robotic actuation under tight timing constraints.

Querétaro, Mexico
Feb 2018 – Mar 2019

SAMSUNG

Education

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| M.Sc. Instituto Tecnológico de Morelia , Digital Electronics
Thesis: “Neural Network Architectures in FPGA/FPSoC”
Developed VHDL auto-generation from TensorFlow-trained networks.
Focused coursework on digital signal processing, ASIC design, and hardware acceleration techniques. | 2019 – 2021 |
| B.Sc. Instituto Tecnológico de Morelia , Electronics Engineering (Embedded Systems) | 2012 – 2018 |

Certifications

- Udemy: Advanced C Programming
- Vision Technologies and Techniques
- High Voltage Safety and PPE
- 7° Encuentro Regional de Energía Renovable
- XXIV Congreso Internacional TECNOTRONICA

Publications

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| “FPGA/FPSoC Neural Networks for Performance-Based Design”
Published at Academia Journals. Presented inference from BNN on Artix-7. | 2021 |
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Languages

Spanish: Native

English: Fluent

German: Basic

Soft Skills

- Agile/Scrum collaboration
- Mentoring junior developers
- Technical decision-making under pressure
- Fast adaptation to legacy codebases
- Clear technical communication across teams

References

Silvia Ramirez – silvia.ramirez@visteon.com 

Dr. Arturo Mendez Patino – arturo.mp@morelia.tecnm.mx 